



Bagging vs Boosting | Key Differences

1. Core Idea

- **Bagging (Bootstrap Aggregating)**
 1. Train **many models in parallel** on **different random subsets** (bootstrapped samples) of the data.
 2. Final prediction = **average (regression)** or **majority vote (classification)**.
 3. Goal: **reduce variance** (stabilize high-variance models like decision trees).
 - **Boosting**
 1. Train **many weak learners sequentially**.
 2. Each new learner focuses more on **mistakes of the previous ones**.
 3. Final prediction = **weighted sum/vote** of learners.
 4. Goal: **reduce bias** and build a **strong model**.
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2. Training Strategy

- **Bagging:** Parallel, independent learners (no dependency).
 - **Boosting:** Sequential, each learner depends on the previous.
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3. Sample Weighting

- **Bagging:** All samples are equally likely (bootstrapping).
 - **Boosting:** Misclassified samples get higher weights → next model focuses on them.
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4. Model Examples

- **Bagging Algorithms:** Random Forest, Bagged Decision Trees.
 - **Boosting Algorithms:** AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost.
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5. Bias-Variance Trade-off

- **Bagging:**
 1. ↓ Variance,
 2. May not reduce bias much.
 - **Boosting:**
 1. ↓ Bias (can achieve very high accuracy),
 2. May ↑ variance if overfitted.
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6. Performance & Use Cases

- **Bagging:** Works well on **high variance models** (e.g., deep trees). More stable & robust on noisy data.

- **Boosting:** Works well when you want **high accuracy** and can tolerate longer training & careful tuning.
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7. Quick Analogy

- **Bagging = "Democracy of independent voters."**
Each voter decides independently, and the majority wins.
 - **Boosting = "Team of experts learning from each other's mistakes."**
Each expert improves based on what the last one missed.
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➤ In summary:

- Use **Bagging** when you want stability & robustness.
 - Use **Boosting** when you want maximum accuracy and can handle complexity.
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